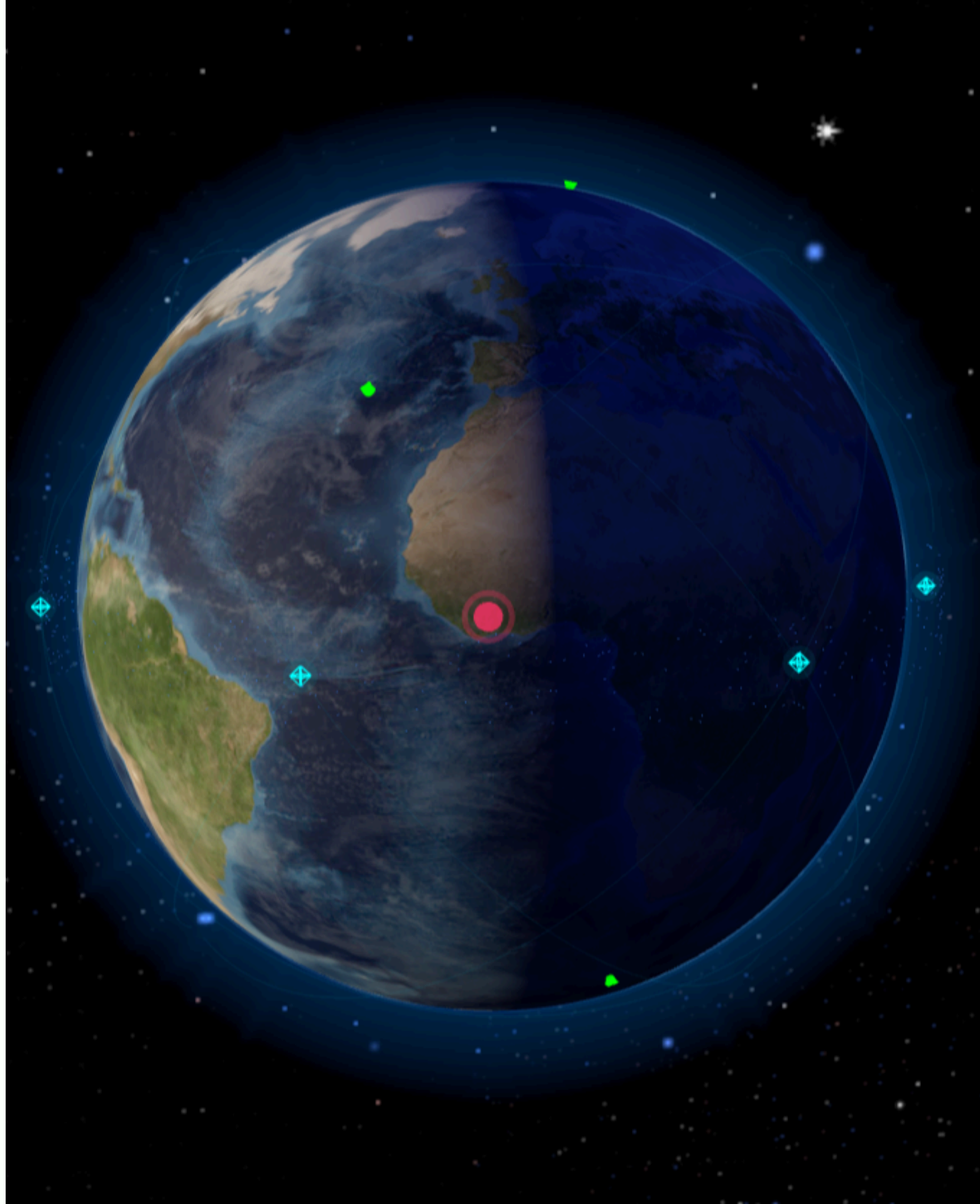
Jawaharlal Nehru Technological University Hyderabad

Orbital Debris Avoidance-Constellation Management System

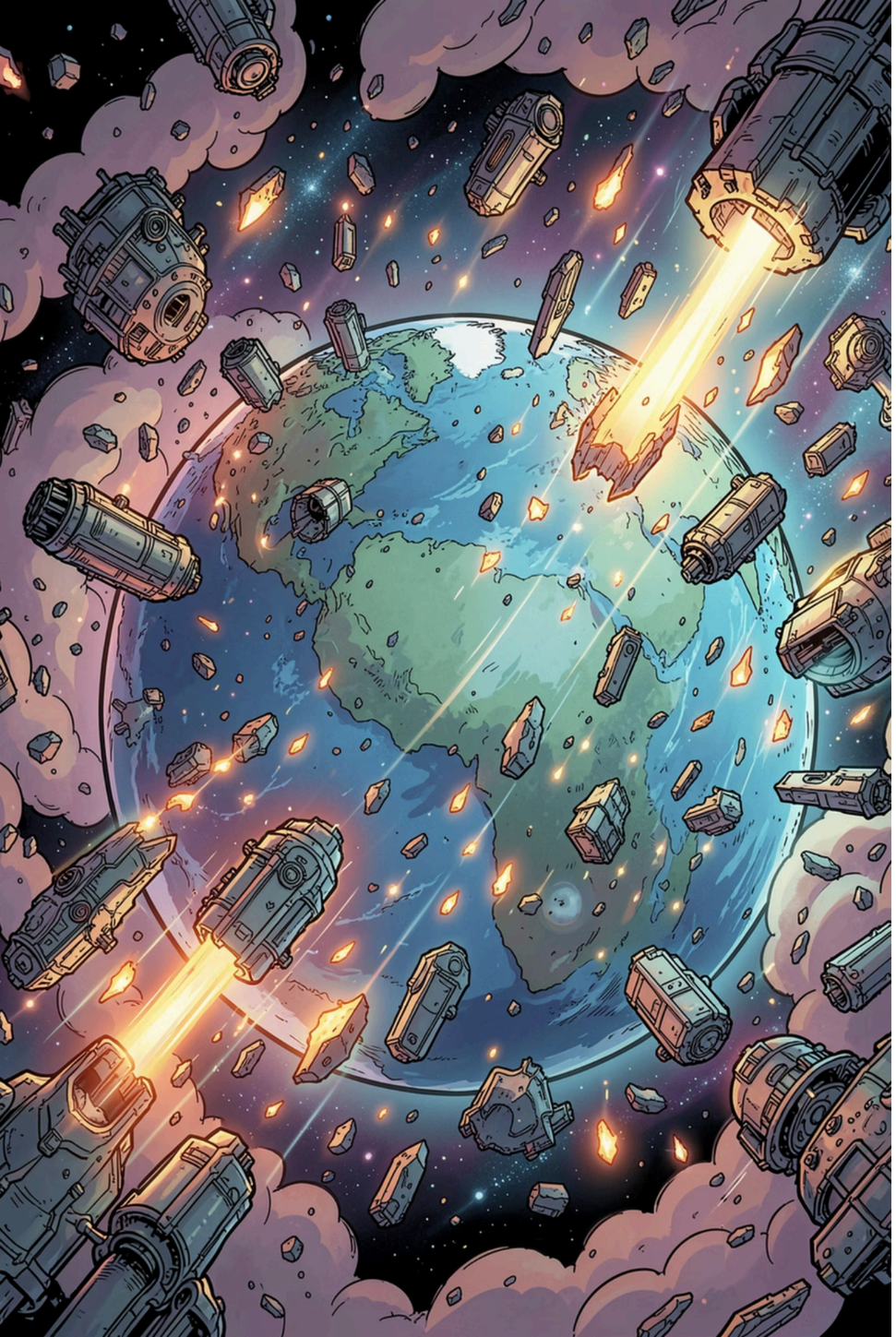
A real-time, intelligent platform for monitoring, predicting, and avoiding space debris collisions — protecting satellite constellations in increasingly congested orbital environments.



Mentor:

G Narasimham

Batch 9



The Growing Threat Above Us

Over **500,000 pieces** of trackable debris orbit Earth at hypersonic speeds. A single collision can trigger a cascade — the **Kessler Syndrome** — rendering entire orbital regions unusable for generations.

500K+

Trackable Objects

Debris fragments tracked by global networks

28,000

km/h Speed

Average orbital velocity — catastrophic on impact

26K+

Active Satellites

Currently at risk in LEO and MEO orbits

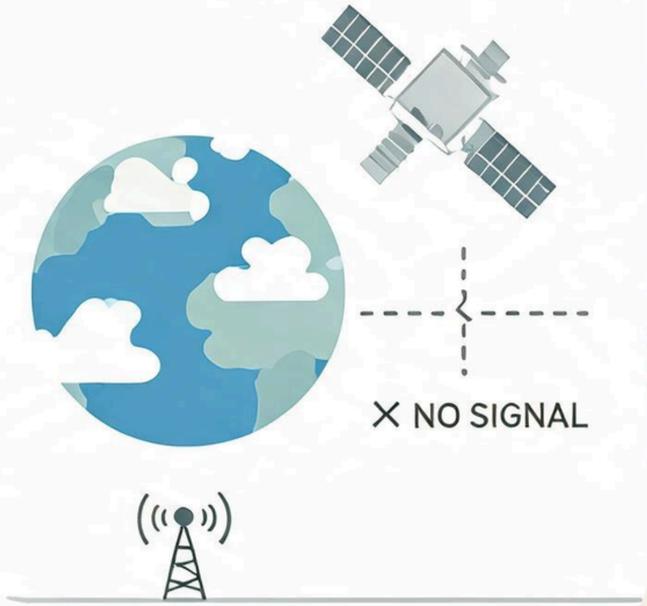
Why Current Systems Fall Short

1. MANUAL CDM REVIEW



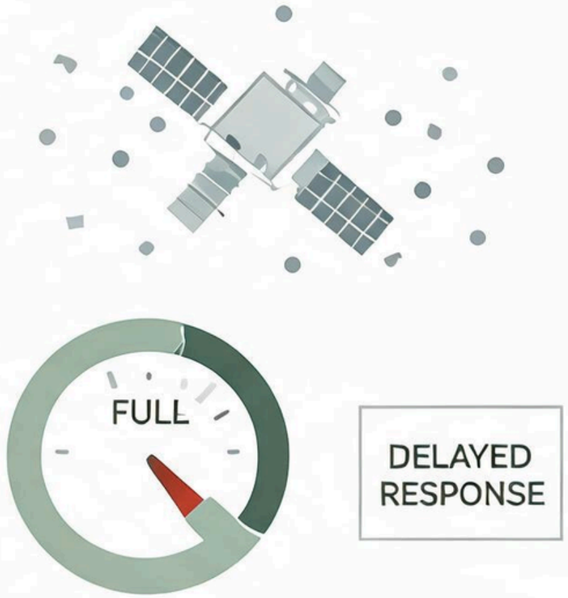
Existing methods depend on manual analysis of Conjunction Data Messages (CDMs), which do not scale for large constellations and frequent alerts.

2. COMMUNICATION BLACKOUT



Satellites can also enter communication blackout zones where ground stations cannot issue timely manoeuvre commands.

3. INEFFICIENT REAL-TIME DECISIONS



Manual operations further struggle to optimize fuel use, recovery burns, and large-scale debris screening in real time.

What Is the ODA-CMS?



The **Orbital Debris Avoidance Constellation Management System** is a full-stack web platform that simulates a 6-satellite constellation, evaluates collision risks, and recommends avoidance maneuvers for satellite operators.

Real-Time Monitoring

Tracks simulated satellites and debris in real time.

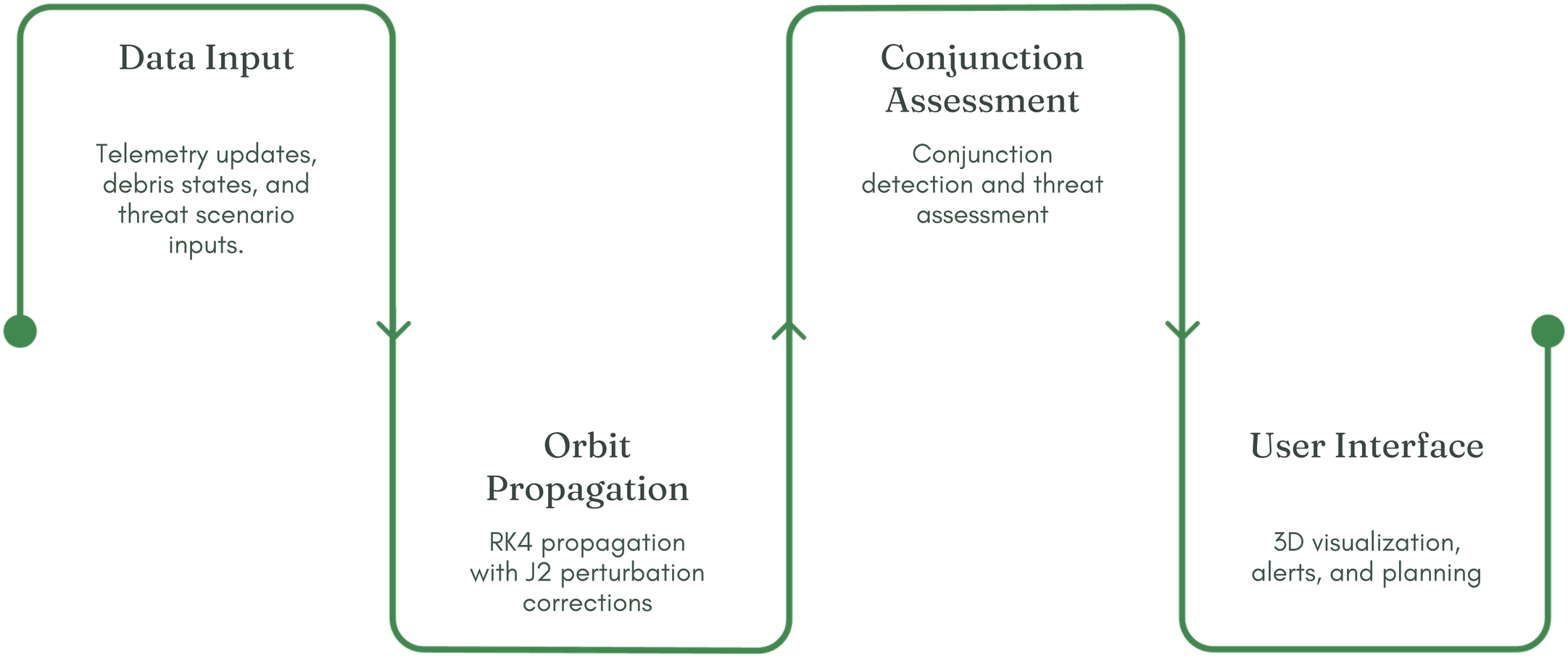
Predictive Analytics

Propagates orbits to detect possible conjunctions.

Automated Alerts

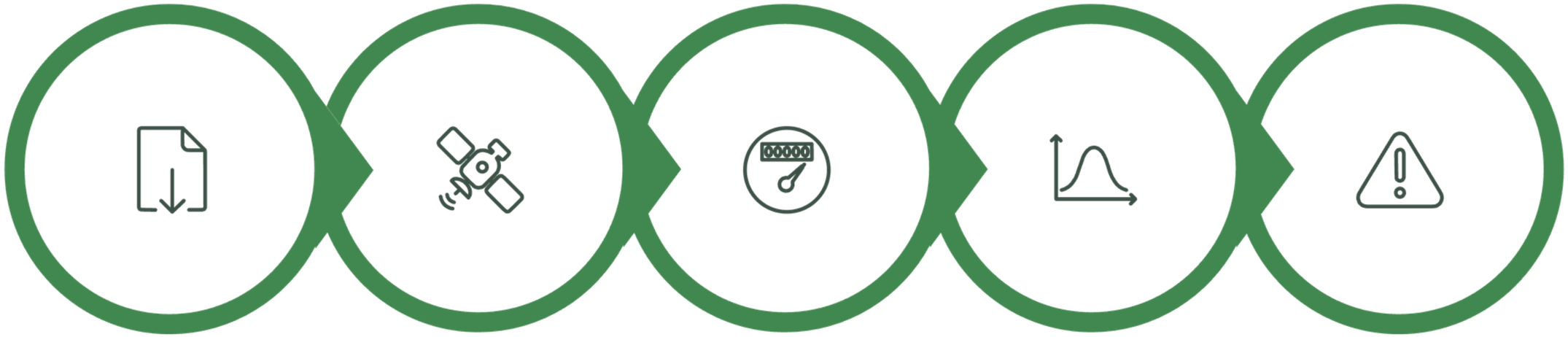
Generates warnings for satellites at risk

System Architecture



The system advances simulated satellite and debris states, detects close-approach threats, schedules avoidance manoeuvres, and presents mission status through an interactive dashboard.

How Collision Avoidance Works



Ingest Data by spawning a threat

Sent to COCA engine

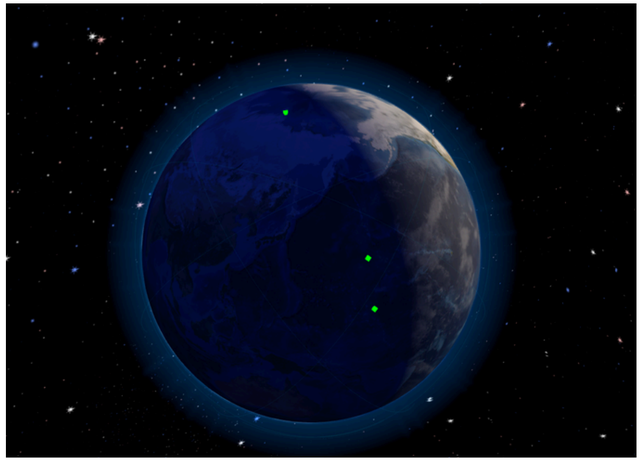
Compute manuever tactic

Plans to manuever Satellite

Trigger Action

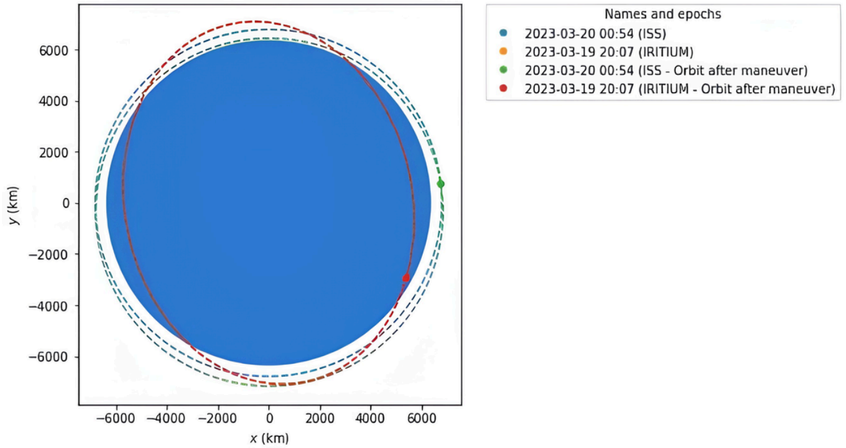
After each close approach is evaluated using propagated object states, miss-distance checks, and warning thresholds to support timely avoidance and recovery decisions

Core Capabilities



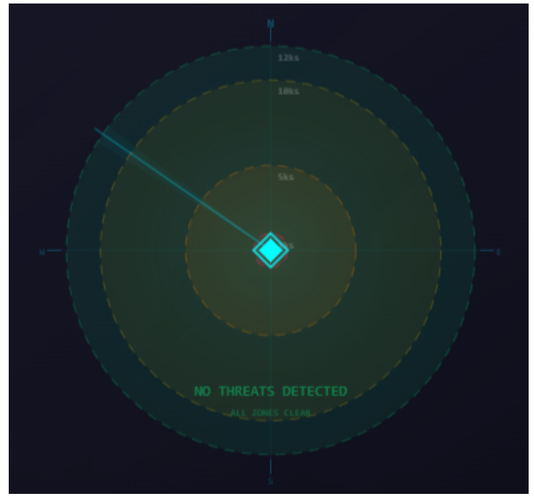
3D Orbital Visualization

Interactive WebGL globe rendering real-time satellite positions, debris clouds, and conjunction geometries.



Autonomous Decision Making

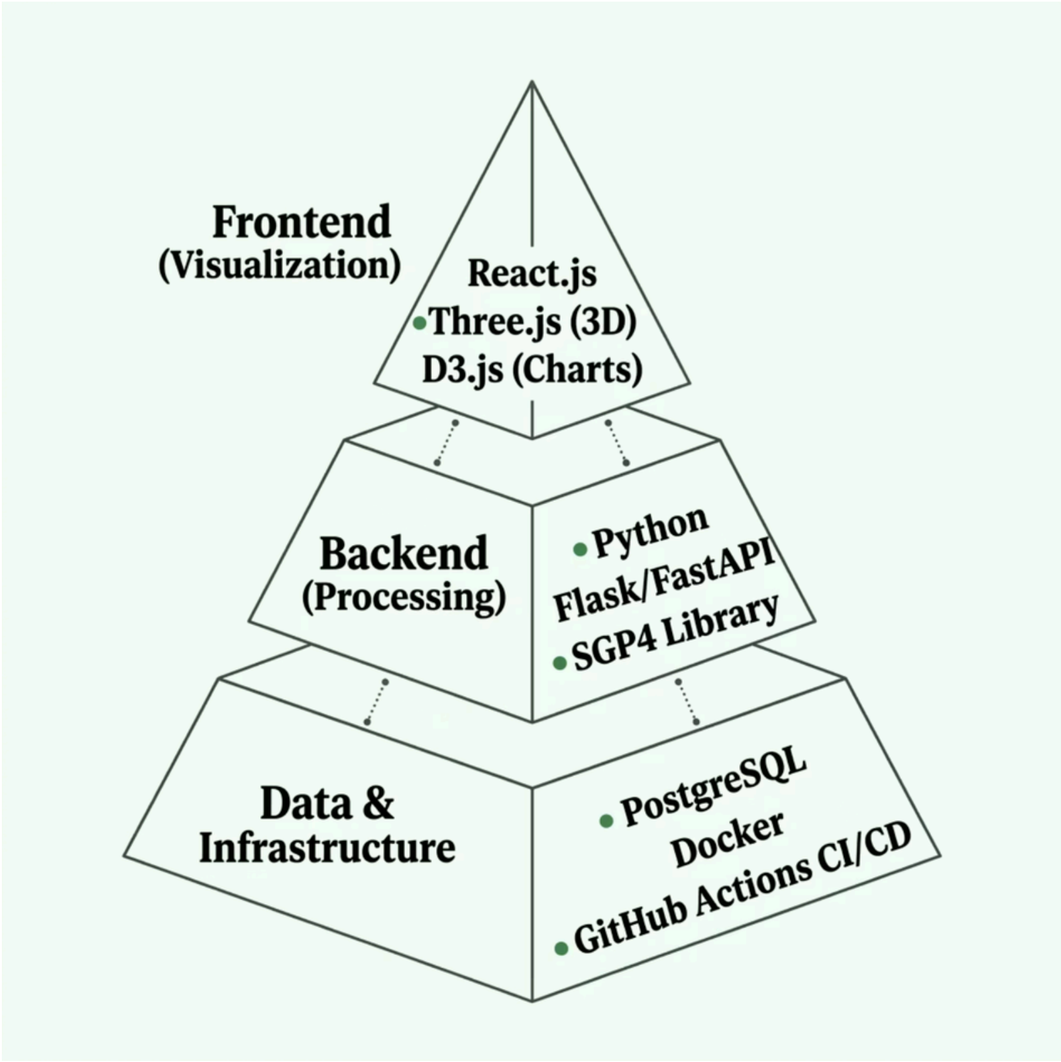
Autonomously makes decision using the COCA engine and manuevers satellite to avoid collision.



Conjunction Threat Detection

Identifies close-approach threats and supports automated avoidance actions.

Technology Stack



Built for Performance and Reliability

The system combines React, FastAPI, and PostgreSQL to deliver responsive visualization, mission logging, and real-time operational awareness

The complete project source code is available on GitHub, with a live frontend deployment available for demonstration and review.



User Experience & Interface

🛰️ Constellation Dashboard

Overview of satellite status, orbital activity, fuel state, and active threats.

🔔 Alert Management

Color-coded warnings with threat details, countdowns, and response visibility.

📊 Historical Analytics

Operational logs for warnings, manoeuvres, fuel usage, and mission events.

Impact & Future Roadmap

Immediate Value

- Reduces operator workload through automated risk screening
- Provides a practical platform for academic and engineering research use.
- Demonstrates end-to-end space situational awareness pipeline

Roadmap Ahead

- Integrate CDM (Conjunction Data Message) format support
- Add spatial indexing for faster large-scale conjunction screening
- Enable coordinated fleet-wide avoidance and recovery planning
- Integrate real orbital data sources and backend container deployment.

Protecting the Orbital Commons

As mega-constellations expand and launch costs fall, intelligent debris management is no longer optional — it is essential infrastructure for sustainable space operations.

🌐 Live Demo

orbital-debris-avoidance-constellat.vercel.app

💻 Source Code

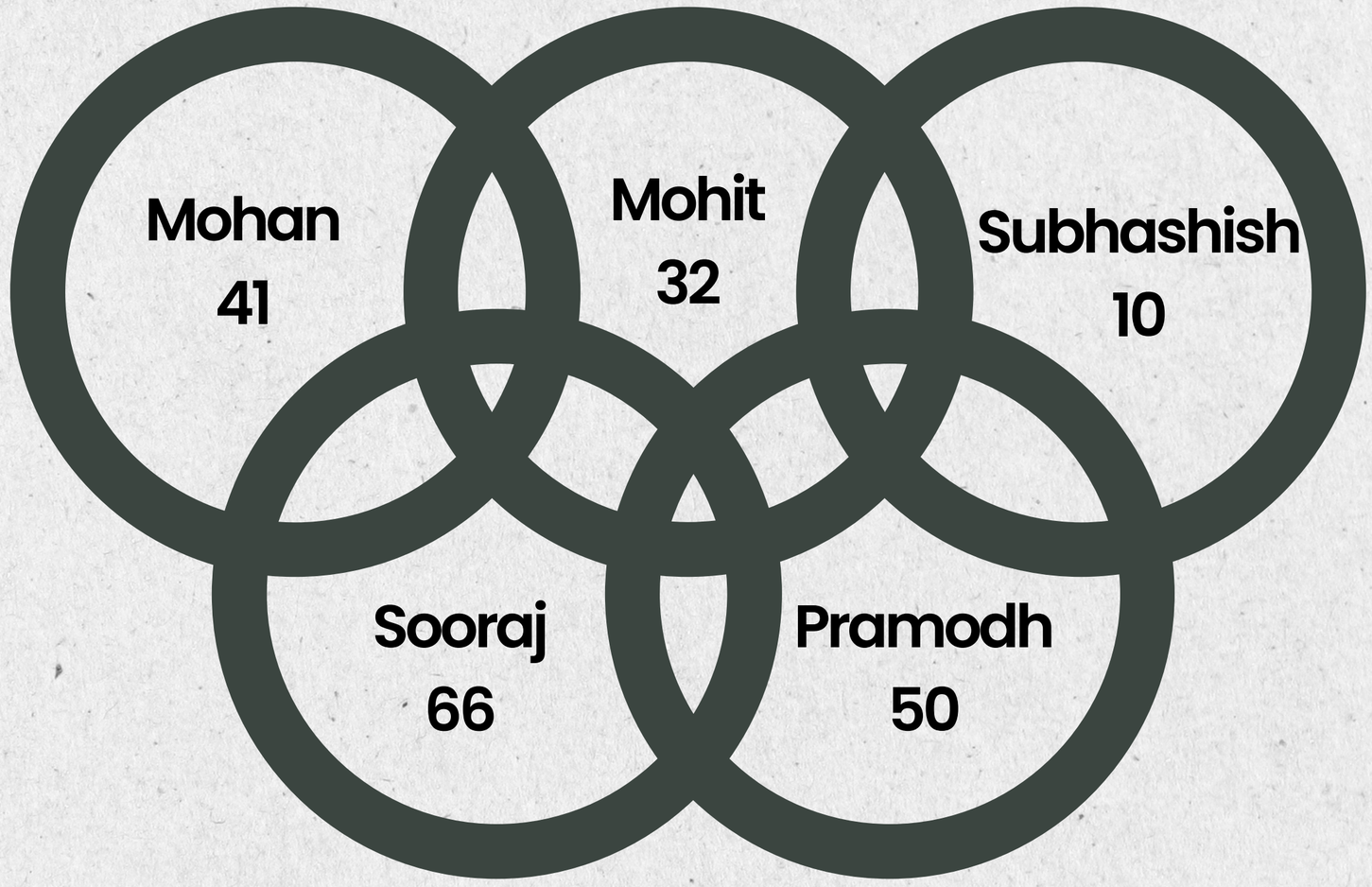
github.com/AridoshikaZu103/Orbital-Debris-Avoidance-Constellation-Management-System

📄 Documentation

Full README, architecture notes, and API reference available on GitHub



MEET THE TEAM



Thank You